THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 5 & 6: **Life Processes, Control and Coordination** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Leaves of plants appear green because chlorophyll: 1

(a) emits green light **(b)** absorbs all light

(c) absorbs green light **(d)** reflects green light

2. A student states: “Plants respire only at night.” The correct response is: 1

(a) Correct **(b)** Correct for desert plants

(c) Incorrect — they respire all the time **(d)** Cannot be determined

3. Fungi such as bread mould obtain food by: 1

(a) saprotrophic nutrition **(b)** parasitic nutrition

(c) photosynthesis **(d)** holozoic nutrition

4. The movement of water in xylem from roots to leaves is mainly due to: 1

(a) gravity **(b)** photosynthesis

(c) transpiration pull **(d)** phloem pressure

5. For blood pressure, Student A says systolic pressure is ~120 mm Hg; Student B says diastolic is ~80 mm Hg. Then: 1

(a) Both are correct **(b)** Only Student B is correct

(c) Only Student A is correct **(d)** Both are wrong

6. Which of these is an involuntary action? 1

(a) walking **(b)** talking **(c)** heartbeat **(d)** writing

7. A student states: “Hormones are secreted by exocrine glands into ducts.” The correct response is: 1

(a) Correct

(b) Only in plants

(c) Incorrect — endocrine glands secrete directly into blood

(d) Cannot be determined

8. The movement of a shoot towards light is: 1

(a) chemotropism **(b)** positive phototropism

(c) positive geotropism **(d)** negative phototropism

9. The brain and spinal cord together form the: 1

(a) autonomic system **(b)** peripheral nervous system

(c) reflex arc **(d)** central nervous system

10. For sugar level control, Student A says insulin lowers blood sugar; Student B says adrenaline lowers it. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the role of HCl in the stomach? 2

12. What would happen if platelets were absent? 2

13. How does the shoot respond to gravity? 2

14. Why do we immediately pull our hand away from a hot object? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How do guard cells regulate gaseous exchange and water loss? 3

16. What are the limitations of electrical impulses in nervous coordination? 3

17. What happens to glucose in muscle cells during vigorous exercise? Why do we get cramps? 3

OR

Why do herbivores have a longer small intestine than carnivores? Give one example of each. 3

18. Distinguish between phototropism and geotropism with examples. 3

OR

How does the touch-me-not plant respond to touch? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A kidney patient undergoes haemodialysis twice a week. 5

(i) What is the principle of dialysis?

(ii) What is removed from the blood?

(iii) How is a nephron similar to an artificial kidney?

(iv) How does it differ?

(v) Name the main waste removed.

OR

(i) Explain the process of digestion of food from mouth to stomach. 5

(ii) What would happen if mucus were not secreted in the stomach?

20. **Case study:** A doctor advised a patient with a swollen neck to use iodised salt. 5

(i) Name the disease.

(ii) Which gland is affected?

(iii) What hormone does this gland produce?

(iv) What is its function?

(v) Why is iodine needed?

OR

(i) Compare nervous and endocrine systems (four points). 5

(ii) Name a gland that is both exocrine and endocrine.

END OF QUESTION PAPER — Total: 40 Marks